MATH70097 - Supervised Learning: Assessment 1

A technology company is running an online advertising campaign. Users are shown a banner while browsing the web. Some users click on the banner and then subscribe to the service. This event is called a conversion. Your task is to predict whether a conversion occurs for each user, based on information about: • the user, • the current advertising campaign, • and a previous related advertising campaign. You will build a predictive model using historical data and use it to make predictions for new users.

Pulling from the lecture notes and assessment brief, we approach this assigment in 4 parts.

1. **Exploratory Data Analysis and Preprocessing** — understand the structure and signals, detect issues (missing values, unusual values, imbalance, nonlinearity).
2. **Building Prediction Models** — train and compare Logistic Regression (with Ridge and Lasso variants), kNN, LDA, QDA, and Naive Bayes.
3. **Validation and Model Selection** — 10-fold stratified CV for hyperparameter tuning, Bootstrap for uncertainty quantification, and principled model selection via log-loss.
4. **Generating Test Predictions** — refit on the full training set and produce predicted probabilities.

Setup & Loading data

We import the relevant packages, set a seed, and load the data. The kernel used is the ISLP (Python 3.12.2) Kernel - which we setup in week 1 of the course.

In [ ]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import StratifiedKFold, cross_val_score, cro
from sklearn.preprocessing import StandardScaler, OneHotEncoder
from sklearn.compose import ColumnTransformer
from sklearn.pipeline import Pipeline
from sklearn.linear_model import LogisticRegression
from sklearn.neighbors import KNeighborsClassifier
from sklearn.discriminant_analysis import LinearDiscriminantAnalysis, Qua
from sklearn.naive_bayes import GaussianNB
from sklearn.metrics import log_loss, confusion_matrix
from sklearn.calibration import calibration_curve
from sklearn.decomposition import PCA
from sklearn.base import clone
import warnings
warnings.filterwarnings('ignore')
```

```
sns.set_theme(style='whitegrid', palette='muted')
pd.set_option('display.max_columns', 25)
SEED = 42
np.random.seed(SEED)
```

In [68]:
```
train = pd.read_csv('train.csv')
test  = pd.read_csv('test.csv')
print(f"Training set : {train.shape[0]} rows, {train.shape[1]} columns")
print(f"Test set     : {test.shape[0]} rows, {test.shape[1]} columns")
train.head()
```

```
Training set : 31647 rows, 17 columns
Test set     : 13564 rows, 17 columns
```

Out[68]:

	age	job	marital	education	X2	X4	X3	X1	device	day	month
0	36	technician	single	tertiary	0	0	0	0	unknown	17	jur
1	56	entrepreneur	married	secondary	0	196	0	0	cellular	19	nov
2	46	blue-collar	married	secondary	0	0	1	0	unknown	5	jur
3	41	management	divorced	tertiary	0	3426	0	0	cellular	1	oct
4	38	blue-collar	married	secondary	0	0	1	0	unknown	20	may

Part 1: Explore the training data (EDA)

Following the data descriptions from the brief, as well as looking through our dataframe, we choose to split the variables into two types. Quantitative variables and Qualitative (categorical) variables.

In [69]:
```
# Define quantitative variables
quantitative = ['age', 'X4', 'day', 'time_spent', 'banner_views', 'days_e

# Define qualitative variables
qualitative = ['job', 'marital', 'education', 'X1', 'X2', 'X3', 'device',

# Print qualitative and quantitative variables

print("Quantitative variables:")
print(train[quantitative].dtypes)
print("\nQualitative variables:")
for col in qualitative:
    levels = train[col].unique()
    print(f"{col} : {len(levels)} levels - {list(levels)}")

train.describe()
```

```
Quantitative variables:
age                   int64
X4                    int64
day                   int64
time_spent            int64
banner_views          int64
days_elapsed_old    float64
banner_views_old      int64
dtype: object

Qualitative variables:
job : 12 levels - ['technician', 'entrepreneur', 'blue-collar', 'managemen
t', 'admin.', 'retired', 'housemaid', 'unknown', 'services', 'student', 'u
nemployed', 'self-employed']
marital : 3 levels - ['single', 'married', 'divorced']
education : 4 levels - ['tertiary', 'secondary', 'primary', 'unknown']
X1 : 2 levels - [np.int64(0), np.int64(1)]
X2 : 2 levels - [np.int64(0), np.int64(1)]
X3 : 2 levels - [np.int64(0), np.int64(1)]
device : 3 levels - ['unknown', 'cellular', 'telephone']
outcome_old : 4 levels - ['unknown', 'success', 'other', 'failure']
month : 12 levels - ['jun', 'nov', 'oct', 'may', 'jul', 'aug', 'feb', 'dec
', 'sep', 'jan', 'mar', 'apr']
```

Out[69]:

	age	X2	X4	X3	X1	
count	31647.000000	31647.000000	31647.000000	31647.000000	31647.000000	3164
mean	40.941669	0.019370	1357.985465	0.556040	0.160458	1
std	10.632010	0.137824	2976.874443	0.496857	0.367036	
min	18.000000	0.000000	-6847.000000	0.000000	0.000000	
25%	33.000000	0.000000	70.000000	0.000000	0.000000	
50%	39.000000	0.000000	446.000000	1.000000	0.000000	1
75%	48.000000	0.000000	1434.000000	1.000000	0.000000	2
max	95.000000	1.000000	81204.000000	1.000000	1.000000	3

We find that there are no null or NaN values in the dataset. However, for 'job', 'education', 'device' and 'outcome_old', we have "unknown" values to serve as a missing indicator.

Similarly, for 'days_elapsed_old', we have values -1, indicating that the user did not see the old banner.

In [70]:
```
print("NaN/Null values in training set:")
print(train.isnull().sum())
```

```
NaN/Null values in training set:
age                 0
job                 0
marital             0
education           0
X2                  0
X4                  0
X3                  0
X1                  0
device              0
day                 0
month               0
time_spent          0
banner_views        0
days_elapsed_old    0
banner_views_old    0
outcome_old         0
y                   0
dtype: int64
```

In [71]:
```
print("unknown values in training set:")
for x in ['job', 'education', 'device', 'outcome_old']:
    unknown_count = (train[x] == 'unknown').sum()
    percentage = (unknown_count / len(train)) * 100
    print(f"{x} : {unknown_count} ({percentage:.2f}%)")
```

```
unknown values in training set:
job : 203 (0.64%)
education : 1322 (4.18%)
device : 9065 (28.64%)
outcome_old : 25867 (81.74%)
```

In [72]:
```
print("No. of days_elapsed_old = -1 in training set:")
print((train['days_elapsed_old'] == -1).sum())

print("No.of days_elapsed_old = -1 in test set:")
print((test['days_elapsed_old'] == -1).sum())
```

```
No. of days_elapsed_old = -1 in training set:
0
No.of days_elapsed_old = -1 in test set:
0
```

Missing values takeaway

"unknown": we can retain "unknown" as a factor level. The lack of a value itself provides us with information on users which were not exposed to the old banner

'device': over 80% of the values are "unknown". We keep the variable but approach it carefully in model development as it carries limited signal to our predictors.

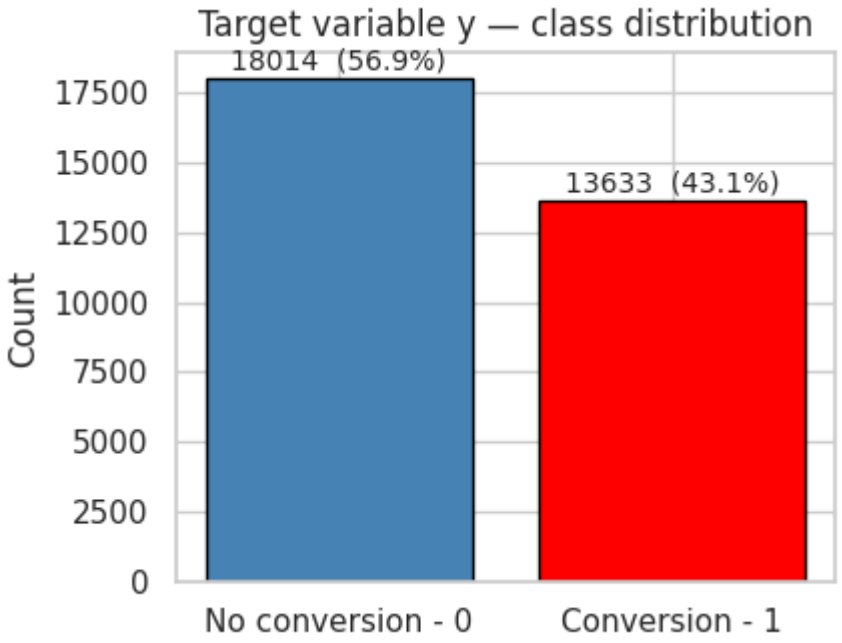
'days_elapsed_old': No -1 values are found in our dataset, so we can avoid handling this case specifically as we don't have any test or train data for it.

Numeric Variables: There are no missing numerical values, so we do not have to perform specific imputation (filling).

After checking for missing and unusual values, we check for imbalances.

In [73]:
```
rate = train['y'].mean()
print(f"Overall conversion rate (y=1) in training set: {rate:.4f}")
ratio = (train['y'] == 0).sum() / (train['y'] == 1).sum()
print(f"Class imbalance ratio (y=0 : y=1) in training set: {ratio:.2f} :

fig, ax = plt.subplots(figsize=(4.5, 3.5))
counts = train['y'].value_counts().sort_index()
bars = ax.bar(['No conversion - 0', 'Conversion - 1'], counts.values,
              color=['steelblue', 'red'], edgecolor='black')
for i, v in enumerate(counts.values):
    ax.text(i, v + 300, f"{v}  ({100*v/len(train):.1f}%)", ha='center', f
ax.set_ylabel('Count')
ax.set_title('Target variable y — class distribution')
plt.tight_layout()
plt.show()
```

```
Overall conversion rate (y=1) in training set: 0.4308
Class imbalance ratio (y=0 : y=1) in training set: 1.32 : 1
```



Imbalances takeaway

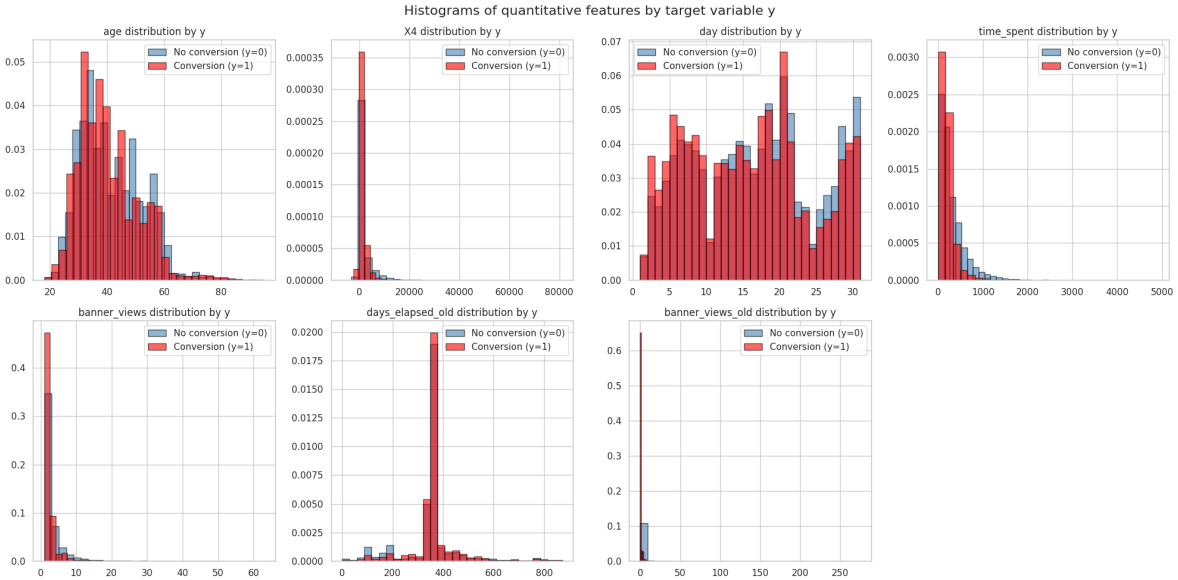
The targeted split has a mild imbalance, 57% non-conversion to 43% conversion. We shouldn't use specific resampling techniques here, but we do focus on a couple of key points:

1. Use stratified cross-validation to preserve class proportions in every fold.
2. Keep the baseline in mind: a naïve classifier predicting the majority class achieves only ~57% accuracy.

We proceed to visualise structure:

In [74]:
```
fig, axes = plt.subplots(2, 4, figsize=(20, 10))
axes = axes.ravel()

for i, col in enumerate(quantitative):
    ax = axes[i]
    for yval, colour, label in [(0, 'steelblue', 'No conversion (y=0)'),
                                (1, 'red', 'Conversion (y=1)')]:
```

```
        subset = train.loc[train['y'] == yval, col]
        ax.hist(subset, bins=30, alpha=0.6, color=colour, label=label, ed
    ax.set_title(f"{col} distribution by y")
    ax.legend()

axes[-1].set_visible(False)
plt.suptitle("Histograms of quantitative features by target variable y",
plt.tight_layout()
plt.show()
```



Quantitative structure takeaways

We find a couple of key observations that may help us out:

X4 is heavily right-skewed, so a log-transform may help with our linear-models.

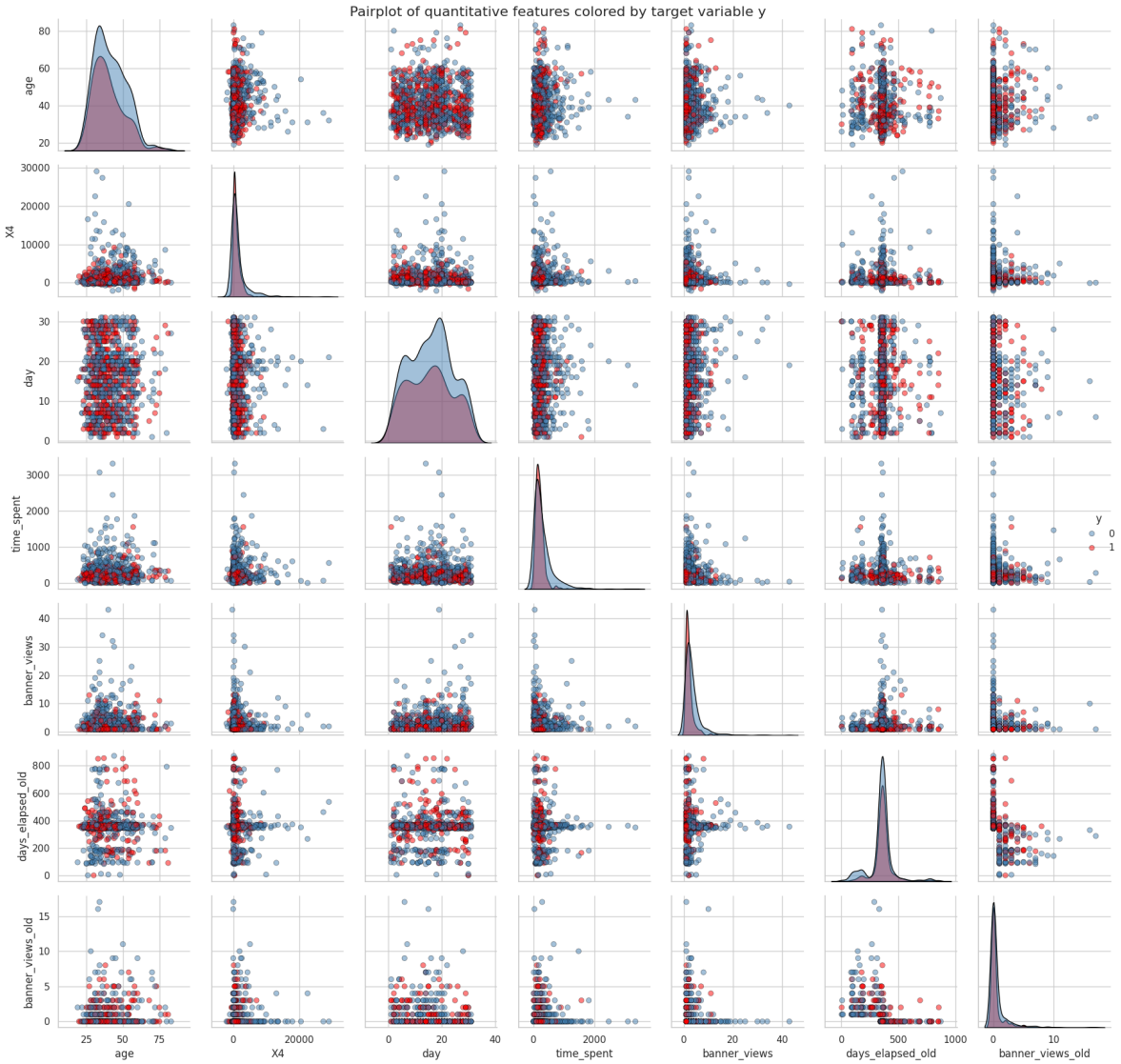
banner_views and banner_views_old are right-skewed.

time_spent interestingly shows a significant shift between classes. Non-conversion (y=0) spend more time. This seems to signal that quick engagement and attention-capture has a larger chance of conversion.

age is (rougly) symmetric, and has some class seperation.

We move onto exploring quantitative variables

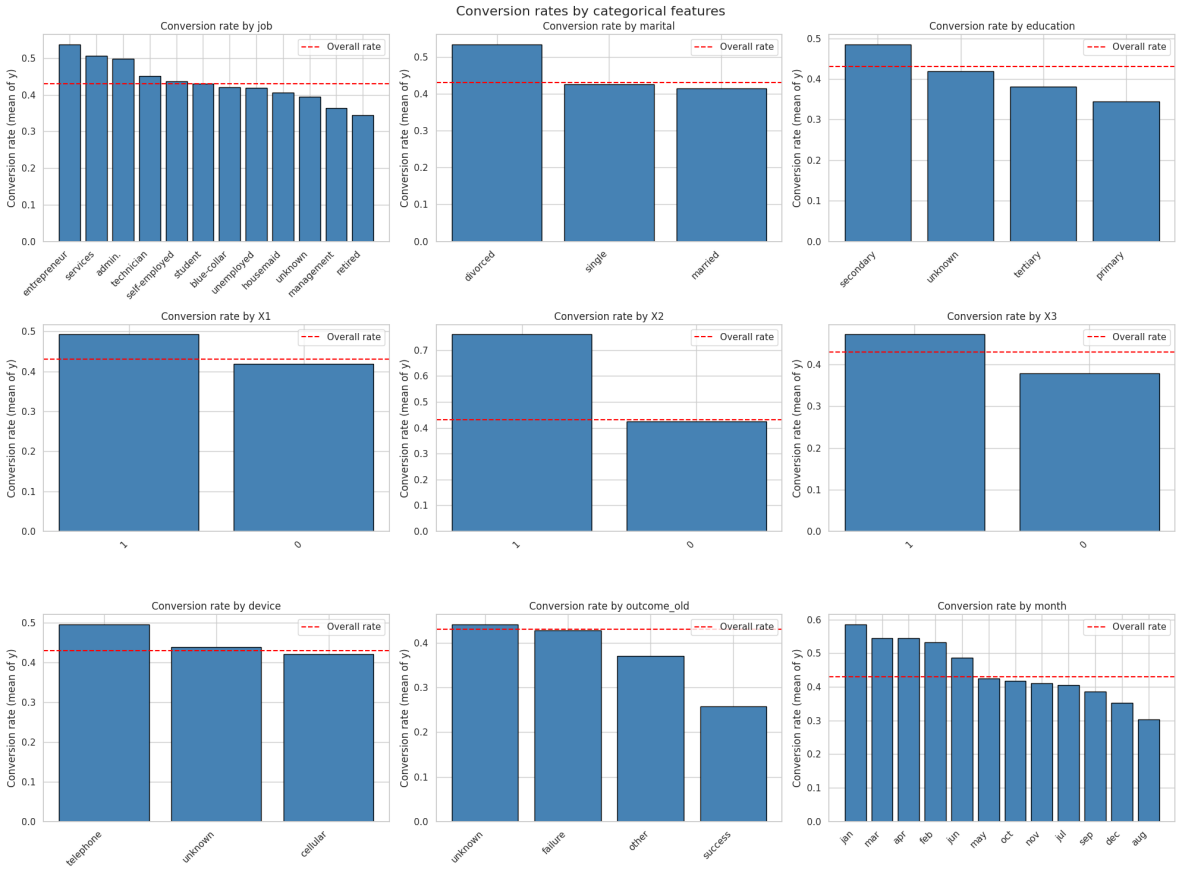
In [75]:
```
vars = quantitative
sample = train[vars + ['y']].sample(1000, random_state=SEED)
sample['y'] = sample['y'].astype(str)

g = sns.pairplot(sample, hue='y', vars=vars, diag_kind='kde',
                 palette={'0': 'steelblue', '1': 'red'},
                 plot_kws={'alpha': 0.5, 'edgecolor': 'black'},
                 diag_kws={'alpha': 0.5, 'edgecolor': 'black'})
g.fig.suptitle("Pairplot of quantitative features colored by target varia
plt.tight_layout()
plt.show()
```



Our pairplot reveals substantial overlaps. No single feature seperates conversion rate cleanly.

We move onto conversion rate by categorical variables

In [76]:
```
cat_vars = qualitative
fig, axes = plt.subplots(3, 3, figsize=(20, 15))
for ax, col in zip(axes.ravel(), cat_vars):
    rates = train.groupby(col)['y'].agg(['mean', 'count']).sort_values('m
    bars = ax.bar(range(len(rates)), rates['mean'], color='steelblue', ed
    ax.set_xticks(range(len(rates)))
    ax.set_xticklabels(rates.index, rotation=45, ha='right')
    ax.set_title(f"Conversion rate by {col}")
    ax.set_ylabel("Conversion rate (mean of y)")
    ax.axhline(train['y'].mean(), color='red', linestyle='--', label='Ove
    ax.legend()
plt.suptitle("Conversion rates by categorical features", fontsize=16)
plt.tight_layout()
plt.show()
```



Key Qualitative (categorical) takeaways:

outcome_old: "success" has the lowest conversion rate, around 26%.

month: suggests seasonal variation, with Jan, Mar, April being the highest.

Job: Students and Retirees convert at the largest rates.

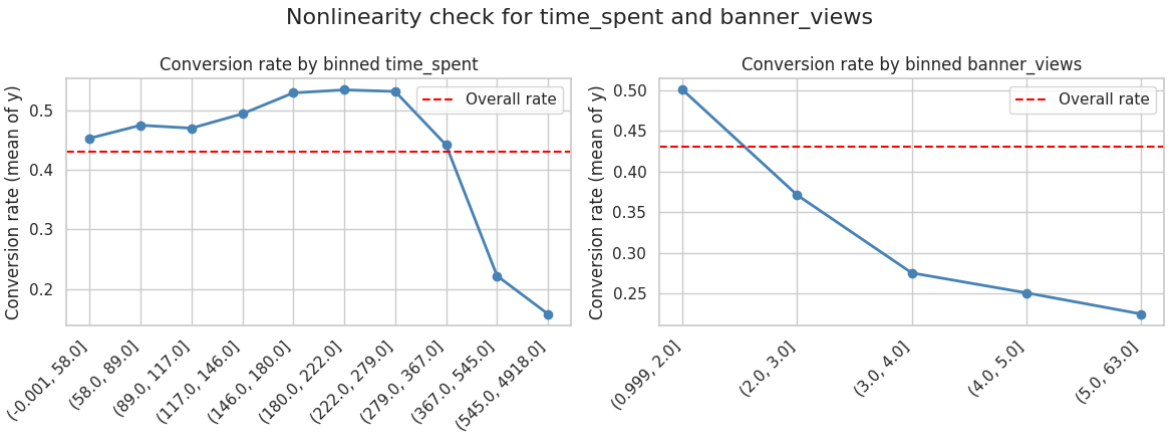
We proceed to check for correlation structure and nonlinerarity

In [77]:
```
# Pearson correlation of quantitative features with target variable y
corr = train[quantitative + ['y']].corr().drop('y').sort_values('y', asce
print("Pearson correlation of target variable y:")
for var, val in corr['y'].items():
    bar = '' * int(abs(val) * 50)
    sign = '+' if val > 0 else '-'
    print(f"{var:20s} : {val:.4f} {sign} {bar}")
```

```
Pearson correlation of target variable y:
time_spent           : -0.2009 -
banner_views         : -0.1687 -
X4                   : -0.0778 -
day                  : -0.0639 -
age                  : -0.0531 -
banner_views_old     : -0.0224 -
days_elapsed_old     : 0.0542 +
```

In [78]:
```
# Nonlinearity check

fig, axes = plt.subplots(1, 2, figsize=(12, 4.5))

for ax, col in zip(axes, ['time_spent', 'banner_views']):
    helper = train.copy()
```

```
    helper[col + '_bin'] = pd.qcut(helper[col], q=10, duplicates='drop')
    rates = helper.groupby(col + '_bin')['y'].mean()
    ax.plot(range(len(rates)), rates.values, 'o-', color='steelblue', lin
    ax.set_xticks(range(len(rates)))
    ax.set_xticklabels(rates.index.astype(str), rotation=45, ha='right')
    ax.set_title(f"Conversion rate by binned {col}")
    ax.set_ylabel("Conversion rate (mean of y)")
    ax.axhline(train['y'].mean(), color='red', linestyle='--', label='Ove
    ax.legend()
plt.suptitle("Nonlinearity check for time_spent and banner_views", fontsi
plt.tight_layout()
plt.show()
```



time_spent and banner_views show decreasing relationships that are monotonic with conversion. This pattern indicates that logistic regression and LDA may be suitable models. On the tails(extremities) we have curvature, which indicates that kNN and QDA may be a reasonable signal capture in the tails.

Dimensionality discussion

The curse of dimensionality is an important theoretical consideration. The number of predictors p is large relative to the structure of the data, expanding our dimensionality.

In [79]:
```
# Dimensionality count after one-hot encoding
var_counts = ['job', 'marital', 'education', 'device', 'outcome_old', 'mo
levels = sum(train[var].nunique() for var in var_counts)

dummies = sum(train[var].nunique() - 1 for var in var_counts)

n_numeric = len(quantitative)
n_binary = 3
n_engineered = 4

p_total = n_numeric + dummies + n_binary + n_engineered
n = len(train)

print(f"Total features after one-hot encoding: {p_total}")
print(f"Number of training samples: {n}")
print(f"n/p ratio: {n/p_total:.0f}")
```

```
Total features after one-hot encoding: 46
Number of training samples: 31647
n/p ratio: 688
```

The n/p ratio is comfortable for parametric models at 688, but the expanded dimensionality has consequences for kNN:

The nearest neighbour degrades. As data points become approximately equidistant, the query point no longer remains local.This leads to kNN bias increasing in average dimensions, as it averages over points that are not close, and so can no longer effectively capture local structure. This leads to a large k to stabilise predictions. We may find that Lasso logistic regression or Naive bayes work better here.

Feature encoding takeways

We define the processing pipeline across all models tested, as such:

Quantitative variables - standardised using StandardScalar for kNN and regularised logistical regression.

Qualitative variables - Converted to dummy variables using one-hot encoding. First level is dropped to avoid multicollinearity, and "unknown" is retained as a useful level.

We proceed with feature engineering:

In [80]:
```
def engineer_features(df):
    df = df.copy()
    df['old_campaign_exposed'] = (df['banner_views_old'] > 0).astype(int)
    df['old_success'] = (df['outcome_old'] == 'success').astype(int)
    df['time_per_view'] = df['time_spent'] / (df['banner_views'] + 1).cli
    df['X4_log'] = np.sign(df['X4']) * np.log1p(np.abs(df['X4']))  # Log-
    return df

train_featured = engineer_features(train)
test_featured = engineer_features(test)

print("New features added: old_campaign_exposed, old_success, time_per_vi
```

```
New features added: old_campaign_exposed, old_success, time_per_view, X4_l
og
```

In [81]:
```
numeric_features = quantitative + ['time_per_view', 'X4_log']
binary_features = ['old_campaign_exposed', 'old_success', 'X1', 'X2', 'X3
categorical_features = ['job', 'marital', 'education', 'device', 'outcome

all_features = numeric_features + binary_features + categorical_features
print(f"Total features after engineering: {len(all_features)}")

x_train = train_featured[all_features]
y_train = train_featured['y']
x_test = test_featured[all_features]
test_ids = test_featured['ID']

# Preprocessor pipeline
preprocessor = ColumnTransformer(
    transformers=[
        ('num', StandardScaler(), numeric_features),
        ('bin', 'passthrough', binary_features),
        ('cat', OneHotEncoder(handle_unknown='ignore', sparse_output=Fals
```

```
    ]
)

print("Preprocessor pipeline created with standard scaling for numeric fe
print(f"feature breakdown: {len(numeric_features)} numeric, {len(binary_f
print(f"X_train shape before preprocessing: {x_train.shape}, X_test shape
```

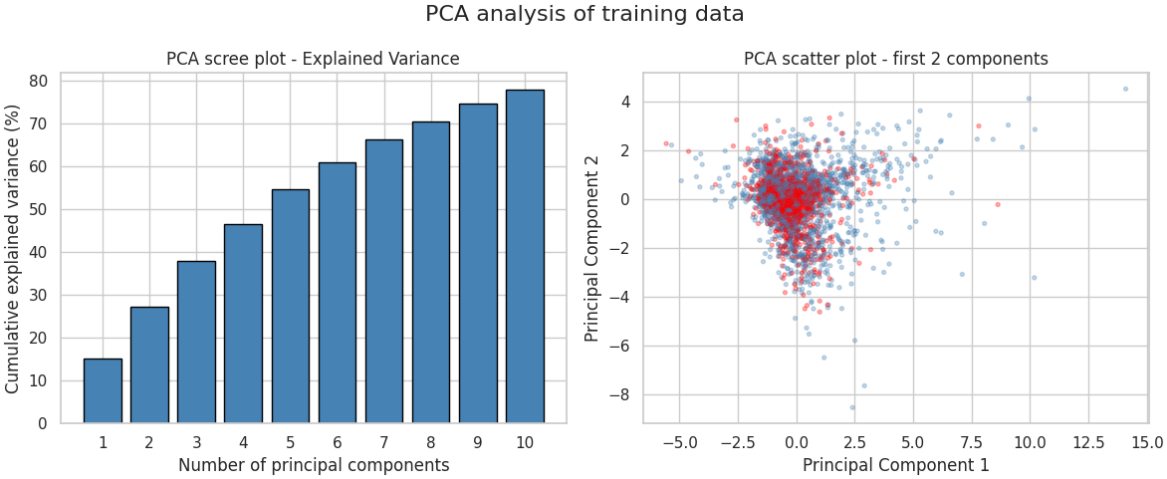
```
Total features after engineering: 20
Preprocessor pipeline created with standard scaling for numeric features,
passthrough for binary features, and one-hot encoding for categorical feat
ures.
feature breakdown: 9 numeric, 5 binary, 6 categorical (one-hot encoded to
32 features)
X_train shape before preprocessing: (31647, 20), X_test shape before prepr
ocessing: (13564, 20)
```

PCA Analysis

PCA Analysis was introduced in week 4 as an unsupervised tool to find maximal variance directions. We apply it to check if the data admits a low-dimensional structure, and visualises class seperation in the reduced spaces.

In [82]:
```
X_train_processed = preprocessor.fit_transform(x_train)
print(f"X_train shape after preprocessing: {X_train_processed.shape}")

pca = PCA(n_components=10, random_state=SEED)
X_pca = pca.fit_transform(X_train_processed)
print(f"Shape of PCA-transformed training data: {X_pca.shape}")

fig, axes = plt.subplots(1, 2, figsize=(12, 5))

ax = axes[0]
cum_var = np.cumsum(pca.explained_variance_ratio_) * 100
ax.bar = ax.bar(range(1, len(cum_var) + 1), cum_var, color='steelblue', e
ax.set_xticks(range(1, len(cum_var) + 1))
ax.set_xticklabels(range(1, len(cum_var) + 1))
ax.set_xlabel('Number of principal components')
ax.set_ylabel('Cumulative explained variance (%)')
ax.set_title('PCA scree plot - Explained Variance')

ax = axes[1]
colours = y_train.map({0: 'steelblue', 1: 'red'})
sample_idx = np.random.choice(len(X_pca), size=3000, replace=False)
ax.scatter(X_pca[sample_idx, 0], X_pca[sample_idx, 1], c=colours.iloc[sam
ax.set_xlabel('Principal Component 1')
ax.set_ylabel('Principal Component 2')
ax.set_title('PCA scatter plot - first 2 components')

plt.suptitle("PCA analysis of training data", fontsize=16)
plt.tight_layout()
plt.show()

print(f"Cumulative variance explained by 5 PCs: {cum_var[4]:.1f}%")
print(f"Cumulative variance explained by 10 PCs: {cum_var[9]:.1f}%")
```

```
X_train shape after preprocessing: (31647, 52)
Shape of PCA-transformed training data: (31647, 10)
```




```
Cumulative variance explained by 5 PCs: 54.5%
Cumulative variance explained by 10 PCs: 77.8%
```

The PCA Scree plot confirms the high-effective dimensionality, there is no clear seperator in low dimensional projections.

EDA Recap

Moderate class imbalance (43% positive): Use stratified CV; log-loss as evaluation metric.

"unknown" encodes structural missingness: Retain as its own category, do not impute.

No literal NaN in numeric columns: No numeric imputation needed.

outcome_old = "success" has the lowest conversion: Strong predictor; include as feature.

time_spent, banner_views negatively correlated with y: Key numeric predictors with a near-linear relationship.

X4 heavily skewed: Apply log-transform.

month, X1 show large group-level differences: Important categorical predictors.

device` mostly unknown: Low signal so we can include but expect low importance.

High effective dimensionality (~40 after one-hot): Curse of dimensionality affects kNN; favours Lasso for variable selection and Naive Bayes for variance reduction.

Pair plot / PCA show overlap, no dramatic nonlinearity: Linear models (LR, LDA) should be competitive

Step 2 - Model Building and Justification

We choose to train 6 models, covered by the course. We see no reason to combine models, as EDA does not strongly suggest a benefit. All models are wrapped in a pipeline with consistent preprocessing. Models are listed as follows:

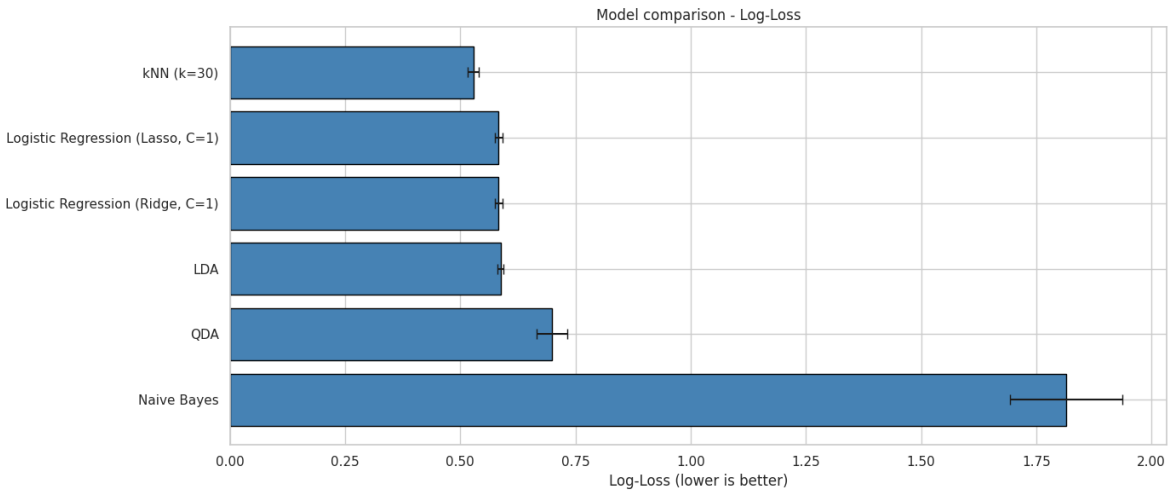
1. Logistic regression (Ride/L2) - directly estimates P(y=1 | X) with L2 regularisation.
2. Logistic regression (Lasso/L1) - same as above but with an L1 penalty, performing automatic variable selection by shrinking irrelevant coeefficients to 0. Useful due to unnamed and dummy variables.
3. k-nearest neighbours - non-parametric, requires standardised features.
4. Linear Discriminant Analysis - assumes each class has a multivariate guassian distribution with a shared covariance matrix.
5. Quadratic Discriminant Analysis - similar to LDA, but allows each class to have its own covariance matrix, leading to flexibility at the cost of more estimate parameters.
6. Naive Bayes - assumes predictors are independent given the class. This "naive" assumption effectively reduces a p-dimensional estimation problem to *p* univariate problems, dramatically reducing variance. This makes it well-suited to the high dimensionality after one-hot encoding, at the cost of potentially increased bias if predictors are correlated.

In [83]:
```
cv = StratifiedKFold(n_splits=10, shuffle=True, random_state=SEED)

models = {
    'Logistic Regression (Ridge, C=1)': Pipeline([
        ('pre', preprocessor),
        ('clf', LogisticRegression(penalty='l2', C=1.0, max_iter=2000,
                                   solver='lbfgs', random_state=SEED))
    ]),
    'Logistic Regression (Lasso, C=1)': Pipeline([
        ('pre', preprocessor),
        ('clf', LogisticRegression(penalty='l1', C=1.0, max_iter=2000,
                                   solver='saga', random_state=SEED))
    ]),
    'kNN (k=30)': Pipeline([
        ('pre', preprocessor),
        ('clf', KNeighborsClassifier(n_neighbors=30))
    ]),
    'LDA': Pipeline([
        ('pre', preprocessor),
        ('clf', LinearDiscriminantAnalysis())
    ]),
    'QDA': Pipeline([
        ('pre', preprocessor),
        ('clf', QuadraticDiscriminantAnalysis(reg_param=0.5))
    ]),
    'Naive Bayes': Pipeline([
        ('pre', preprocessor),
        ('clf', GaussianNB())
    ]),
}

print(f"{'Model':<40s}  {'Log-Loss':>18s}  {'ROC-AUC':>18s}")
print('-' * 80)

results = {}
for name, pipe in models.items():
    ll_scores  = cross_val_score(pipe, x_train, y_train, cv=cv,
                                 scoring='neg_log_loss', n_jobs=-1)
```

```
    auc_scores = cross_val_score(pipe, x_train, y_train, cv=cv,
                                 scoring='roc_auc', n_jobs=-1)
    results[name] = {
        'log_loss_mean': -ll_scores.mean(),
        'log_loss_std':   ll_scores.std(),
        'auc_mean':       auc_scores.mean(),
        'auc_std':        auc_scores.std(),
    }
    print(f"{name:<40s}  {-ll_scores.mean():.4f} ± {ll_scores.std():.4f}
          f"{auc_scores.mean():.4f} ± {auc_scores.std():.4f}")
```

```
Model                                                Log-Loss            R
OC-AUC
------------------------------------------------------------------------
------
Logistic Regression (Ridge, C=1)          0.5834 ± 0.0085   0.7597 ± 0.010
0
Logistic Regression (Lasso, C=1)          0.5833 ± 0.0085   0.7598 ± 0.010
0
kNN (k=30)                                0.5281 ± 0.0123   0.8174 ± 0.010
1
LDA                                       0.5872 ± 0.0071   0.7524 ± 0.010
2
QDA                                       0.6983 ± 0.0334   0.7682 ± 0.010
8
Naive Bayes                               1.8148 ± 0.1212   0.7059 ± 0.011
5
```

In [86]:
```
res_df = pd.DataFrame(results).T.sort_values('log_loss_mean')

fig, ax = plt.subplots(figsize=(14, 6))
ax.barh(res_df.index, res_df['log_loss_mean'], xerr=res_df['log_loss_std'
        color='steelblue', edgecolor='black', capsize=4)
ax.set_xlabel('Log-Loss (lower is better)')
ax.set_title('Model comparison - Log-Loss')
ax.invert_yaxis()
plt.tight_layout()
plt.show()
```



Initial comparison takeaways:

1. Ridge and Lasso Logistic regression perform similarly at C = 1. We will tune for C for each seperately.
2. LDA compares to logistic regression, as expected as we estimate similar linear

boundaries.
3. Naive Bayes has a significant log-loss, indicating that bias may be introduced as predictors are correlated (time_spent, banner views)
4. QDA seems to struggle with high-dimensional one-hot encoded features
5. KNN has not been tuned yet - its performance depends on k.

Step 3 - Validation and model selection

We use a 10-fold stratified cross-validation to tune hyperparameters, with the primary metric being log-loss to select models. We also use Bootstrao to quantify uncertainty for our performance metrics and assess whether differences are statistically meaningful.

Logistic Regression - Ridge tuning:

In [87]:
```
C_values = [0.001, 0.01, 0.05, 0.1, 0.5, 1.0, 5.0, 10.0, 50.0, 100.0]
ridge_results = []

for C in C_values:
    pipe = Pipeline([
        ('pre', preprocessor),
        ('clf', LogisticRegression(penalty='l2', C=C, max_iter=2000,
                                   solver='lbfgs', random_state=SEED))
    ])
    scores = cross_val_score(pipe, x_train, y_train, cv=cv,
                             scoring='neg_log_loss', n_jobs=-1)
    ridge_results.append({
        'C': C,
        'log_loss_mean': -scores.mean(),
        'log_loss_std': scores.std()
    })
    print(f"Ridge Logistic Regression (C={C}) - log-loss: {-scores.mean()

ridge_df = pd.DataFrame(ridge_results)
best_C_ridge = ridge_df.loc[ridge_df['log_loss_mean'].idxmin(), 'C']
best_ll_ridge = ridge_df['log_loss_mean'].min()
print(f"\nBest Ridge Logistic Regression - C={best_C_ridge}, log-loss = {
```

```
Ridge Logistic Regression (C=0.001) - log-loss: 0.6036 ± 0.0061
Ridge Logistic Regression (C=0.01) - log-loss: 0.5863 ± 0.0081
Ridge Logistic Regression (C=0.05) - log-loss: 0.5836 ± 0.0084
Ridge Logistic Regression (C=0.1) - log-loss: 0.5834 ± 0.0085
Ridge Logistic Regression (C=0.5) - log-loss: 0.5834 ± 0.0085
Ridge Logistic Regression (C=1.0) - log-loss: 0.5834 ± 0.0085
Ridge Logistic Regression (C=5.0) - log-loss: 0.5834 ± 0.0085
Ridge Logistic Regression (C=10.0) - log-loss: 0.5834 ± 0.0085
Ridge Logistic Regression (C=50.0) - log-loss: 0.5834 ± 0.0085
Ridge Logistic Regression (C=100.0) - log-loss: 0.5834 ± 0.0085

Best Ridge Logistic Regression - C=0.5, log-loss = 0.5834
```

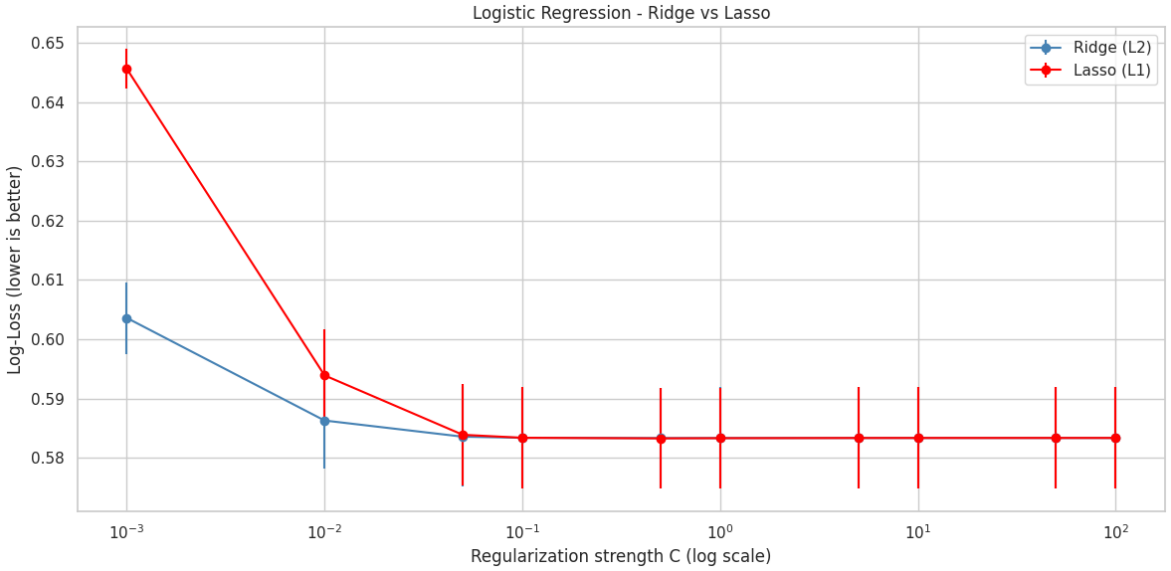
Logistic Regression - Lasso Tuning:

In [88]:
```
lasso_results = []

for C in C_values:
```

```
    pipe = Pipeline([
        ('pre', preprocessor),
        ('clf', LogisticRegression(penalty='l1', C=C, max_iter=2000,
                                   solver='saga', random_state=SEED))
    ])
    scores = cross_val_score(pipe, x_train, y_train, cv=cv, scoring='neg_
    lasso_results.append({
        'C': C,
        'log_loss_mean': -scores.mean(),
        'log_loss_std': scores.std()
    })
    print(f"Lasso Logistic Regression (C={C}) - log-loss: {-scores.mean()

lasso_df = pd.DataFrame(lasso_results)
best_C_lasso = lasso_df.loc[lasso_df['log_loss_mean'].idxmin(), 'C']
best_ll_lasso = lasso_df['log_loss_mean'].min()
print(f"\nBest Lasso Logistic Regression - C={best_C_lasso}, log-loss = {
```

```
Lasso Logistic Regression (C=0.001) - log-loss: 0.6456 ± 0.0033
Lasso Logistic Regression (C=0.01) - log-loss: 0.5939 ± 0.0077
Lasso Logistic Regression (C=0.05) - log-loss: 0.5839 ± 0.0085
Lasso Logistic Regression (C=0.1) - log-loss: 0.5834 ± 0.0085
Lasso Logistic Regression (C=0.5) - log-loss: 0.5833 ± 0.0085
Lasso Logistic Regression (C=1.0) - log-loss: 0.5833 ± 0.0085
Lasso Logistic Regression (C=5.0) - log-loss: 0.5834 ± 0.0085
Lasso Logistic Regression (C=10.0) - log-loss: 0.5834 ± 0.0085
Lasso Logistic Regression (C=50.0) - log-loss: 0.5834 ± 0.0085
Lasso Logistic Regression (C=100.0) - log-loss: 0.5834 ± 0.0085

Best Lasso Logistic Regression - C=0.5, log-loss = 0.5833
```

In [90]:
```
# Compare Ridge and Lasso results

fig, ax = plt.subplots(figsize=(12, 6))
ax.errorbar(ridge_df['C'], ridge_df['log_loss_mean'], yerr=ridge_df['log_
ax.errorbar(lasso_df['C'], lasso_df['log_loss_mean'], yerr=lasso_df['log_
ax.set_xscale('log')
ax.set_xlabel('Regularization strength C (log scale)')
ax.set_ylabel('Log-Loss (lower is better)')
ax.set_title('Logistic Regression - Ridge vs Lasso')
ax.legend()
plt.tight_layout()
plt.show()
```



Model does not seem overtly sensitive to regularisation. If the best log-loss results are similar, Lasso is preferred for its variable selection properties, leading to a more interpretable model.

We continue to check which cofefficients shrunk to 0 for the best C for the Lasso model.

In [ ]:
```
# Fit lasso on best C on full training data
lasso_inspect = Pipeline([
    ('pre', preprocessor),
    ('clf', LogisticRegression(penalty='l1', C=best_C_lasso, max_iter=200
                               solver='saga', random_state=SEED))
])
lasso_inspect.fit(x_train, y_train)

# Get feature names after preprocessing
ohe = lasso_inspect.named_steps['pre'].named_transformers_['cat']
cat_feature_names = ohe.get_feature_names_out(categorical_features).tolis
feature_names = numeric_features + binary_features + cat_feature_names

lasso_coefs = pd.Series(lasso_inspect.named_steps['clf'].coef_[0], index=
n_zero = (lasso_coefs == 0).sum()
n_nonzero = (lasso_coefs != 0).sum()

print(f"Total features after preprocessing: {len(feature_names)}")
print(f"Number of features with zero coefficients: {n_zero}")
print(f"Number of features with non-zero coefficients: {n_nonzero}")
print(("Features removed by Lasso (zero coefficients):\n")
      + "\n".join(lasso_coefs[lasso_coefs == 0].index.tolist()))

print("Survival of features with non-zero coefficients in Lasso:")
surviving_features = lasso_coefs[lasso_coefs != 0].reindex(
    lasso_coefs[lasso_coefs != 0].abs().sort_values(ascending=False).inde
for feat, coef in surviving_features.items():
    print(f"{feat:40s} : {coef:.4f}")
```

```
Total features after preprocessing: 52
Number of features with zero coefficients: 3
Number of features with non-zero coefficients: 49
Features removed by Lasso (zero coefficients):
job_technician
job_unknown
month_oct
Survival of features with non-zero coefficients in Lasso:
X2                                      : 2.2372
time_spent                              : -0.9016
banner_views                            : -0.5902
month_mar                               : 0.5135
month_jan                               : 0.5109
month_aug                               : -0.5068
X3                                      : 0.4761
month_nov                               : -0.4630
X4_log                                  : 0.4339
device_telephone                        : 0.4264
job_entrepreneur                        : 0.4245
device_unknown                          : -0.4130
X4                                      : -0.4089
education_primary                       : -0.3984
old_campaign_exposed                    : -0.3894
marital_divorced                        : 0.3762
month_may                               : -0.3602
month_jul                               : -0.3426
marital_single                          : -0.2997
X1                                      : 0.2773
month_apr                               : 0.2668
job_management                          : -0.2541
month_jun                               : 0.2395
job_retired                             : -0.2221
old_success                             : -0.2107
outcome_old_success                     : -0.2107
time_per_view                           : 0.2081
education_secondary                     : 0.2079
marital_married                         : -0.1905
outcome_old_other                       : -0.1537
month_feb                               : 0.1453
job_student                             : -0.1413
outcome_old_unknown                     : 0.1353
job_unemployed                          : -0.1329
device_cellular                         : -0.1273
job_services                            : 0.1235
age                                     : -0.1093
job_blue-collar                         : -0.0859
day                                     : -0.0763
education_tertiary                      : -0.0714
days_elapsed_old                        : 0.0694
education_unknown                       : 0.0417
month_dec                               : -0.0399
outcome_old_failure                     : 0.0231
banner_views_old                        : 0.0213
month_sep                               : -0.0189
job_self-employed                       : 0.0148
job_admin.                              : 0.0136
job_housemaid                           : 0.0093
```

Tuning kNN

The choice of k directly controls the bias-variance trade-off:

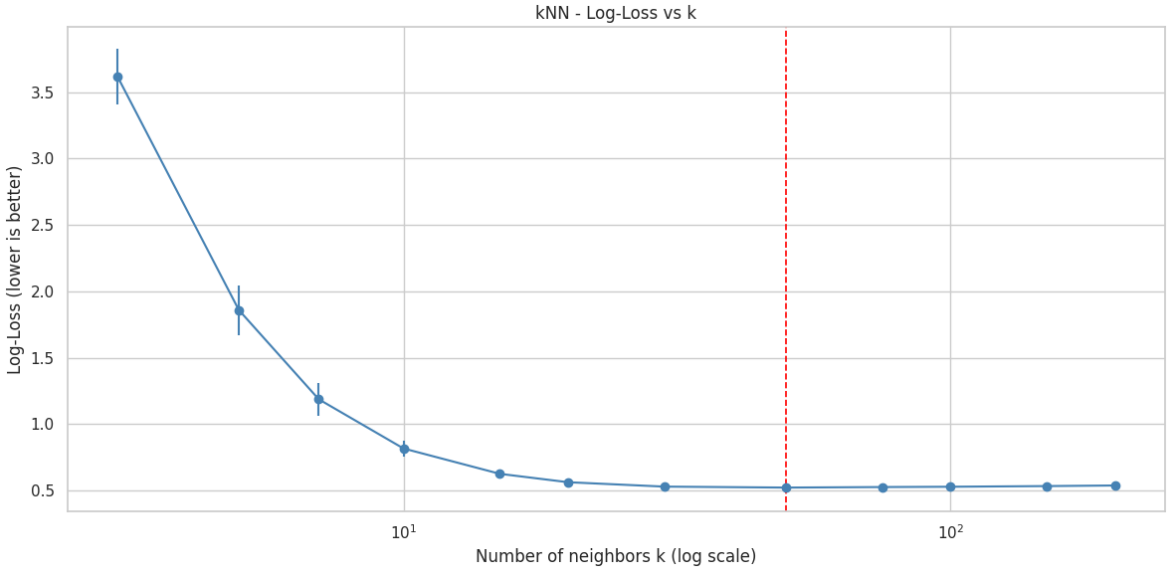
- Small k leads to low bias, high variance (overfitting risk).
- Large k leads to high bias, low variance (underfitting risk).

Given the curse of dimensionality discussion in Section 1.5, we expect kNN to need a relatively large k in this ~50-dimensional feature space, because the neighbourhoods are no longer truly "local" — all points are approximately equidistant in high dimensions.

In [94]:
```
k_values = [3, 5, 7, 10, 15, 20, 30, 50, 75, 100, 150, 200]

knn_results = []
for k in k_values:
    pipe = Pipeline([
        ('pre', preprocessor),
        ('clf', KNeighborsClassifier(n_neighbors=k))
    ])
    scores = cross_val_score(pipe, x_train, y_train, cv=cv,
                             scoring='neg_log_loss', n_jobs=-1)
    knn_results.append({
        'k': k,
        'log_loss_mean': -scores.mean(),
        'log_loss_std': scores.std()
    })
    print(f"kNN (k={k}) - log-loss: {-scores.mean():.4f} ± {scores.std():

knn_df = pd.DataFrame(knn_results)
best_k = knn_df.loc[knn_df['log_loss_mean'].idxmin(), 'k']
best_ll_knn = knn_df['log_loss_mean'].min()
print(f"\nBest kNN - k={best_k}, log-loss = {best_ll_knn:.4f}")
```

```
kNN (k=3) - log-loss: 3.6183 ± 0.2066
kNN (k=5) - log-loss: 1.8586 ± 0.1874
kNN (k=7) - log-loss: 1.1861 ± 0.1256
kNN (k=10) - log-loss: 0.8155 ± 0.0607
kNN (k=15) - log-loss: 0.6244 ± 0.0211
kNN (k=20) - log-loss: 0.5612 ± 0.0215
kNN (k=30) - log-loss: 0.5281 ± 0.0123
kNN (k=50) - log-loss: 0.5208 ± 0.0099
kNN (k=75) - log-loss: 0.5247 ± 0.0091
kNN (k=100) - log-loss: 0.5270 ± 0.0072
kNN (k=150) - log-loss: 0.5321 ± 0.0064
kNN (k=200) - log-loss: 0.5366 ± 0.0065

Best kNN - k=50, log-loss = 0.5208
```

In [100…
```
fig, ax = plt.subplots(figsize=(12, 6))
ax.errorbar(knn_df['k'], knn_df['log_loss_mean'], yerr=knn_df['log_loss_s
ax.set_xscale('log')
ax.axvline(best_k, ls='--', color='red', lw=1.2, label=f'Best k = {best_k
ax.set_xlabel('Number of neighbors k (log scale)')
ax.set_ylabel('Log-Loss (lower is better)')
ax.set_title('kNN - Log-Loss vs k')
plt.tight_layout()
plt.show()
```



As anticipated, the optimal K is relatively large. Small k overfits, due to high variance from noisy estimates, while large k underfits. This shows us that the large dimensional space isn't truely local, and the model averages over many points to get stable estimates. This negates the flexibility that makes kNN attractive in lower dimensions.

LDA and QDA

In [102…

```
# LDA (no hyperparameters to tune) - just evaluate log-loss
lda_pipe = Pipeline([
    ('pre', preprocessor),
    ('clf', LinearDiscriminantAnalysis())
])
lda_scores = cross_val_score(lda_pipe, x_train, y_train, cv=cv, scoring='
lda_log_loss_mean = -lda_scores.mean()
lda_log_loss_std = lda_scores.std()
print(f"LDA - log-loss: {lda_log_loss_mean:.4f} ± {lda_log_loss_std:.4f}"

# QDA (with reg_param tuning - start from 0.1 to avoid singular covarianc
reg_params = [0.1, 0.25, 0.5, 0.75, 0.9]
qda_results = []

for reg in reg_params:
    pipe = Pipeline([
        ('pre', preprocessor),
        ('clf', QuadraticDiscriminantAnalysis(reg_param=reg))
    ])
    scores = cross_val_score(pipe, x_train, y_train, cv=cv, scoring='neg_
    qda_results.append({
        'reg_param': reg,
        'log_loss_mean': -scores.mean(),
        'log_loss_std': scores.std()
    })
    print(f"QDA (reg_param={reg}) - log-loss: {-scores.mean():.4f} ± {sco

qda_df = pd.DataFrame(qda_results)
best_reg = qda_df.loc[qda_df['log_loss_mean'].idxmin(), 'reg_param']
best_ll_qda = qda_df['log_loss_mean'].min()
print(f"\nBest QDA - reg_param={best_reg}, log-loss = {best_ll_qda:.4f}")
```

```
if lda_log_loss_mean < best_ll_qda:
    print("LDA outperforms QDA. The extra flexibility of QDA is not justi
else:
    print("QDA with regularization outperforms LDA.")
```

```
LDA - log-loss: 0.5872 ± 0.0071
QDA (reg_param=0.1) - log-loss: 0.8836 ± 0.0681
QDA (reg_param=0.25) - log-loss: 0.7880 ± 0.0510
QDA (reg_param=0.5) - log-loss: 0.6983 ± 0.0334
QDA (reg_param=0.75) - log-loss: 0.6442 ± 0.0205
QDA (reg_param=0.9) - log-loss: 0.6257 ± 0.0119

Best QDA - reg_param=0.9, log-loss = 0.6257
LDA outperforms QDA. The extra flexibility of QDA is not justified here.
```

Naive Bayes

In [103…

```
# Naive bayes with variance smoothing search
var_smoothing_values = [1e-12, 1e-11, 1e-10, 1e-9, 1e-8, 1e-7, 1e-6, 1e-5
nb_results = []
for var_smooth in var_smoothing_values:
    pipe = Pipeline([
        ('pre', preprocessor),
        ('clf', GaussianNB(var_smoothing=var_smooth))
    ])
    scores = cross_val_score(pipe, x_train, y_train, cv=cv, scoring='neg_
    nb_results.append({
        'var_smoothing': var_smooth,
        'log_loss_mean': -scores.mean(),
        'log_loss_std': scores.std()
    })
    print(f"Naive Bayes (var_smoothing={var_smooth}) - log-loss: {-scores

nb_df = pd.DataFrame(nb_results)
best_var_smooth = nb_df.loc[nb_df['log_loss_mean'].idxmin(), 'var_smoothi
best_ll_nb = nb_df['log_loss_mean'].min()
print(f"\nBest Naive Bayes - var_smoothing={best_var_smooth}, log-loss =
```

```
Naive Bayes (var_smoothing=1e-12) - log-loss: 1.8148 ± 0.1212
Naive Bayes (var_smoothing=1e-11) - log-loss: 1.8148 ± 0.1212
Naive Bayes (var_smoothing=1e-10) - log-loss: 1.8148 ± 0.1212
Naive Bayes (var_smoothing=1e-09) - log-loss: 1.8148 ± 0.1212
Naive Bayes (var_smoothing=1e-08) - log-loss: 1.8148 ± 0.1212
Naive Bayes (var_smoothing=1e-07) - log-loss: 1.8148 ± 0.1212
Naive Bayes (var_smoothing=1e-06) - log-loss: 1.8148 ± 0.1212
Naive Bayes (var_smoothing=1e-05) - log-loss: 1.8141 ± 0.1212
Naive Bayes (var_smoothing=0.0001) - log-loss: 1.8075 ± 0.1203
Naive Bayes (var_smoothing=0.001) - log-loss: 1.7430 ± 0.1169
Naive Bayes (var_smoothing=0.01) - log-loss: 1.3596 ± 0.0917
Naive Bayes (var_smoothing=0.1) - log-loss: 1.0024 ± 0.0701
Naive Bayes (var_smoothing=1.0) - log-loss: 0.7534 ± 0.0261

Best Naive Bayes - var_smoothing=1.0, log-loss = 0.7534
```

Comparison of all tuned models:

In [105…

```
tuned_models = {
    f"ridge LR (C={best_C_ridge})": Pipeline([
        ('pre', preprocessor),
        ('clf', LogisticRegression(penalty='l2', C=best_C_ridge, max_iter
```

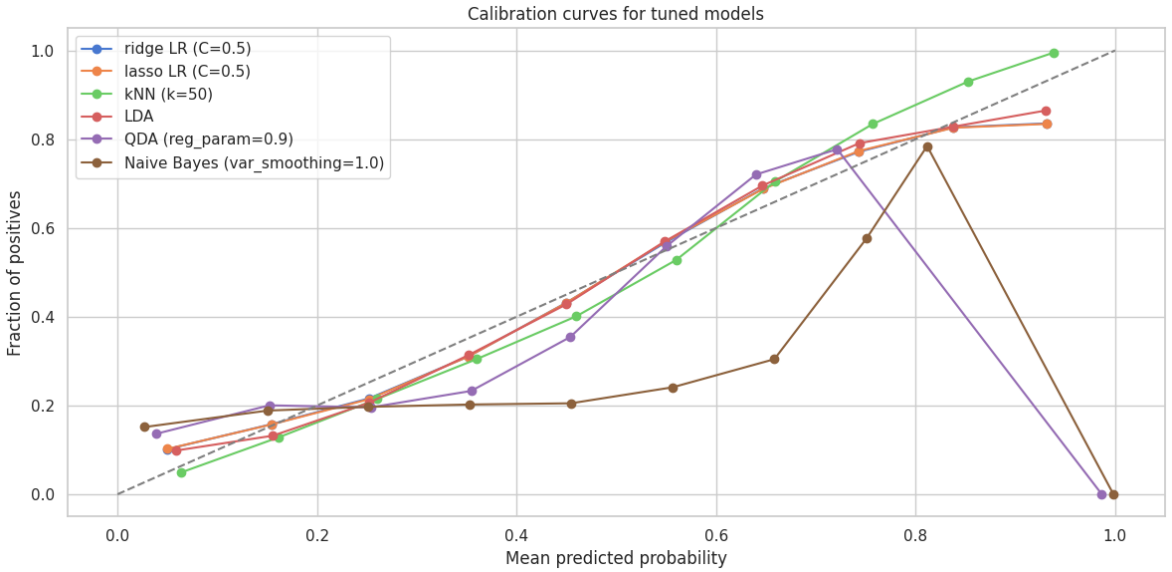
```
                                    solver='lbfgs', random_state=SEED))
    ]),
    f"lasso LR (C={best_C_lasso})": Pipeline([
        ('pre', preprocessor),
        ('clf', LogisticRegression(penalty='l1', C=best_C_lasso, max_iter
                                    solver='saga', random_state=SEED))
    ]),
    f"kNN (k={best_k})": Pipeline([
        ('pre', preprocessor),
        ('clf', KNeighborsClassifier(n_neighbors=best_k))
    ]),
    "LDA": Pipeline([
        ('pre', preprocessor),
        ('clf', LinearDiscriminantAnalysis())
    ]),
    f"QDA (reg_param={best_reg})": Pipeline([
        ('pre', preprocessor),
        ('clf', QuadraticDiscriminantAnalysis(reg_param=best_reg))
    ]),
    f"Naive Bayes (var_smoothing={best_var_smooth})": Pipeline([
        ('pre', preprocessor),
        ('clf', GaussianNB(var_smoothing=best_var_smooth))
    ]),
}

print(f"{'Model':<40s}  {'Log-Loss':>18s}")
print('-' * 60)

final_results = {}
for name, pipe in tuned_models.items():
    ll_scores  = cross_val_score(pipe, x_train, y_train, cv=cv,
                                 scoring='neg_log_loss', n_jobs=-1)
    final_results[name] = {
        'log_loss_mean': -ll_scores.mean(),
        'log_loss_std':   ll_scores.std(),
    }
    print(f"{name:<40s}  {-ll_scores.mean():.4f} ± {ll_scores.std():.4f}"

    best_model = min(final_results, key=lambda x: final_results[x]['log_l
print(f"\nBest overall model: {best_model} with log-loss = {final_results
```

```
Model                                                 Log-Loss
------------------------------------------------------------
ridge LR (C=0.5)                                 0.5834 ± 0.0085
lasso LR (C=0.5)                                 0.5833 ± 0.0085
kNN (k=50)                                       0.5208 ± 0.0099
LDA                                              0.5872 ± 0.0071
QDA (reg_param=0.9)                              0.6257 ± 0.0119
Naive Bayes (var_smoothing=1.0)                  0.7534 ± 0.0261

Best overall model: kNN (k=50) with log-loss = 0.5208 ± 0.0099
```

In [106…

```
# Calibration curves

fig, axes = plt.subplots(figsize=(12, 6))
for name, pipe in tuned_models.items():
    y_prob = cross_val_predict(pipe, x_train, y_train, cv=cv, method='pre
    prob_true, prob_pred = calibration_curve(y_train, y_prob, n_bins=10)
    axes.plot(prob_pred, prob_true, marker='o', label=name)
axes.plot([0, 1], [0, 1], ls='--', color='gray')
```

```
axes.set_xlabel('Mean predicted probability')
axes.set_ylabel('Fraction of positives')
axes.set_title('Calibration curves for tuned models')
axes.legend()
plt.tight_layout()
plt.show()
```



Calibration analysis takeaways

1. Logistic Regression (both Ridge and Lasso) typically produces well-calibrated probabilities by construction — the logistic function naturally outputs values in [0, 1] that are trained via maximum likelihood.
2. kNN probabilities are inherently coarse (they can only take values that are multiples of 1/k), which can lead to poor log-loss even if discrimination is acceptable.
3. Naive Bayes may produce overconfident probabilities (pushed towards 0 or 1) due to the independence assumption which compounds evidence more aggressively than warranted when predictors are correlated.
4. LDA/QDA calibration depends on how well the Gaussian assumptions hold

Bootstrapping

We now use bootstrapping to quantify the uncertainty in our estimates. We estimate the standard error by drawing B samples with the replacement from the training data, fitting the models, and evaluating on out-of-bag observations

In [109…

```
def bootstrap_log_loss(pipe, X, y, B=200, seed = 42):
    rng = np.random.RandomState(seed)
    n = len(X)

    # Reset index so iloc and positional rng.choice stay aligned
    X = X.reset_index(drop=True)
    y = y.reset_index(drop=True)

    boot_losses = []

    for b in range(B):
```

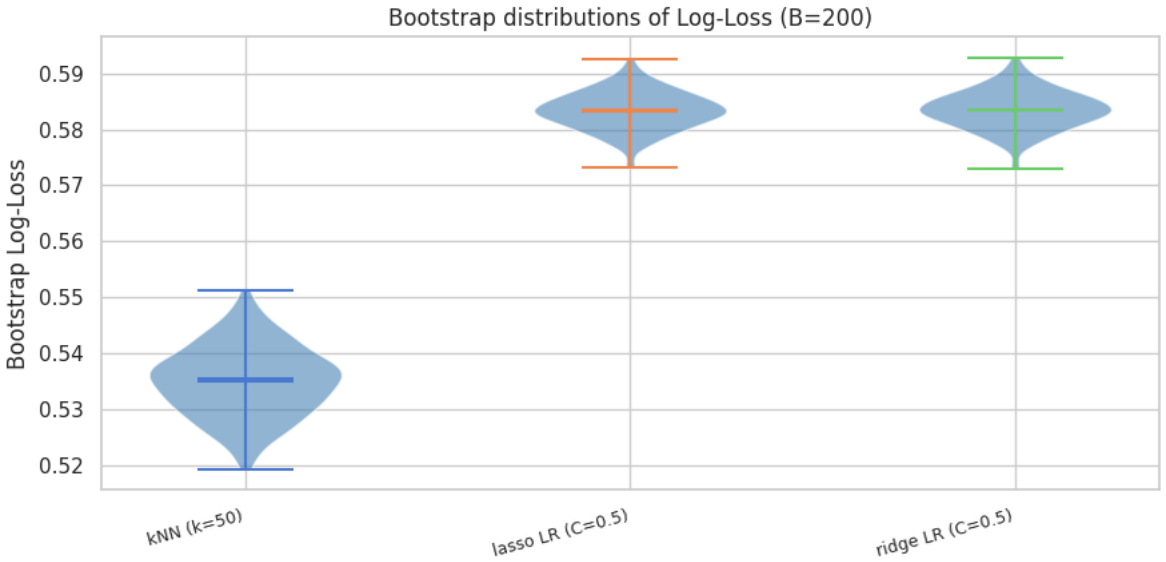
```
        idx_boot = rng.choice(n, size=n, replace=True)
        idx_oob = np.setdiff1d(range(n), idx_boot)
        if len(idx_oob) < 10:
            continue
        X_boot, y_boot = X.iloc[idx_boot], y.iloc[idx_boot]
        X_oob, y_oob = X.iloc[idx_oob], y.iloc[idx_oob]
        if len(y_oob.unique()) < 2 or len(y_boot.unique()) < 2:
            continue
        try:
            pipe_clone = clone(pipe)

            clf = pipe_clone.named_steps.get('clf')
            if clf is not None and hasattr(clf, 'solver'):
                clf.max_iter = 5000
                clf.tol = 1e-3
                clf.n_jobs = 1 # Avoid nested parallelism

            pipe_clone.fit(X_boot, y_boot)
            y_oob_prob = pipe_clone.predict_proba(X_oob)[:, 1]
            boot_losses.append(log_loss(y_oob, y_oob_prob))

        except Exception as e:
            print(f"Bootstrap iteration {b} failed: {e}")
            continue

    return np.array(boot_losses)

# Run bootstrap for best 3 models by log-loss
top_models = sorted(final_results.items(), key=lambda x: x[1]['log_loss_m
bootstrap_results = {}
for name, _ in top_models:
    print(f"Running bootstrap for {name}...")
    pipe = tuned_models[name]
    boot_losses = bootstrap_log_loss(pipe, x_train, y_train, B=200, seed=
    bootstrap_results[name] = boot_losses
    print(f"{name} - Bootstrap log-loss: mean={boot_losses.mean():.4f}, s
```

```
Running bootstrap for kNN (k=50)...
kNN (k=50) - Bootstrap log-loss: mean=0.5351, std=0.0062
Running bootstrap for lasso LR (C=0.5)...
lasso LR (C=0.5) - Bootstrap log-loss: mean=0.5834, std=0.0034
Running bootstrap for ridge LR (C=0.5)...
ridge LR (C=0.5) - Bootstrap log-loss: mean=0.5836, std=0.0035
```

In [113…

```
# Visualise bootstrap distributions
fig, ax = plt.subplots(figsize=(9, 4.5))

for i, (name, boot_ll) in enumerate(bootstrap_results.items()):
    parts = ax.violinplot(boot_ll, positions=[i], showmeans=True, showmed
    for pc in parts['bodies']:
        pc.set_facecolor(colours[i])
        pc.set_alpha(0.6)

ax.set_xticks(range(len(bootstrap_results)))
ax.set_xticklabels(bootstrap_results.keys(), fontsize=9, rotation=15, ha=
ax.set_ylabel('Bootstrap Log-Loss')
ax.set_title('Bootstrap distributions of Log-Loss (B=200)')
plt.tight_layout()
plt.show()
```

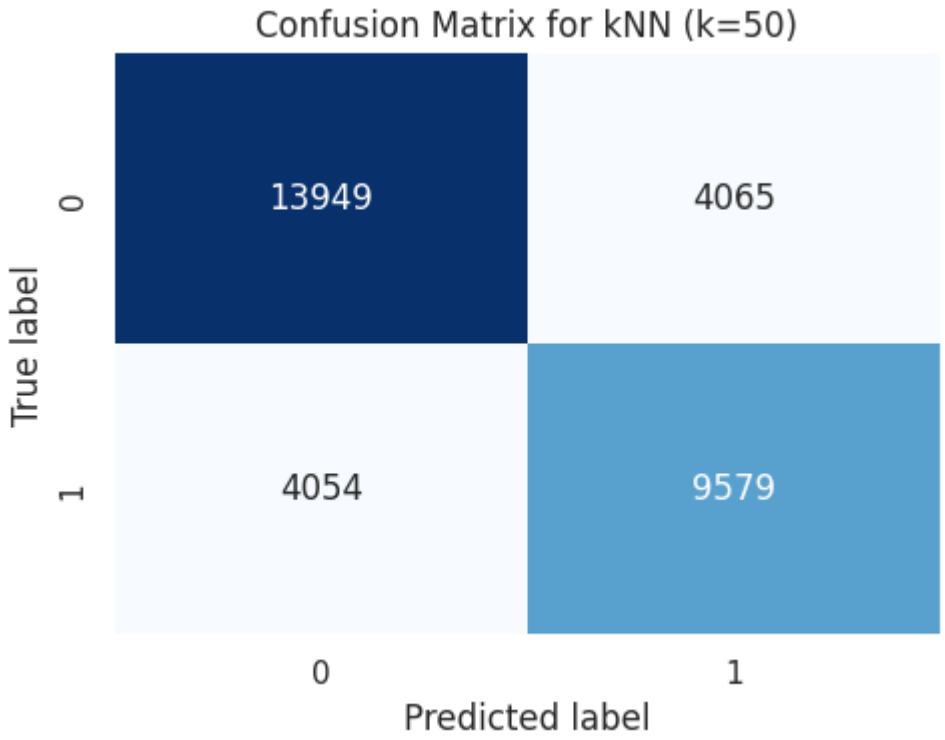
```
# Report on overlap
names = list(bootstrap_results.keys())
if len(names) >= 2:
    ll_1 = bootstrap_results[names[0]]
    ll_2 = bootstrap_results[names[1]]
    diff = ll_2 - ll_1[:len(ll_2)]  # element-wise difference
    pct_better = (diff > 0).mean() * 100
    print(f"\nIn {pct_better:.0f}% of bootstrap samples, {names[0]} has l
    print(f"Mean difference: {diff.mean():.5f} ± {diff.std():.5f}")
    if diff.mean() - 1.96 * diff.std() > 0:
        print(f"The difference is statistically significant at the 95% co
    else:
        print(f"The difference is not statistically significant at the 95
```




```
In 100% of bootstrap samples, kNN (k=50) has lower log-loss than lasso LR
(C=0.5).
Mean difference: 0.04836 ± 0.00598
The difference is statistically significant at the 95% confidence level.
```

In [115…

```
# Confusion matrix for best model

best_pipe = tuned_models[best_model]
prob_cv = cross_val_predict(best_pipe, x_train, y_train, cv=cv, method='p
pred_cv = (prob_cv >= 0.5).astype(int)

cm = confusion_matrix(y_train, pred_cv)
fig, ax = plt.subplots(figsize=(5, 4))
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues', cbar=False, ax=ax)
ax.set_xlabel('Predicted label')
ax.set_ylabel('True label')
ax.set_title(f'Confusion Matrix for {best_model}')
plt.tight_layout()
plt.show()
```



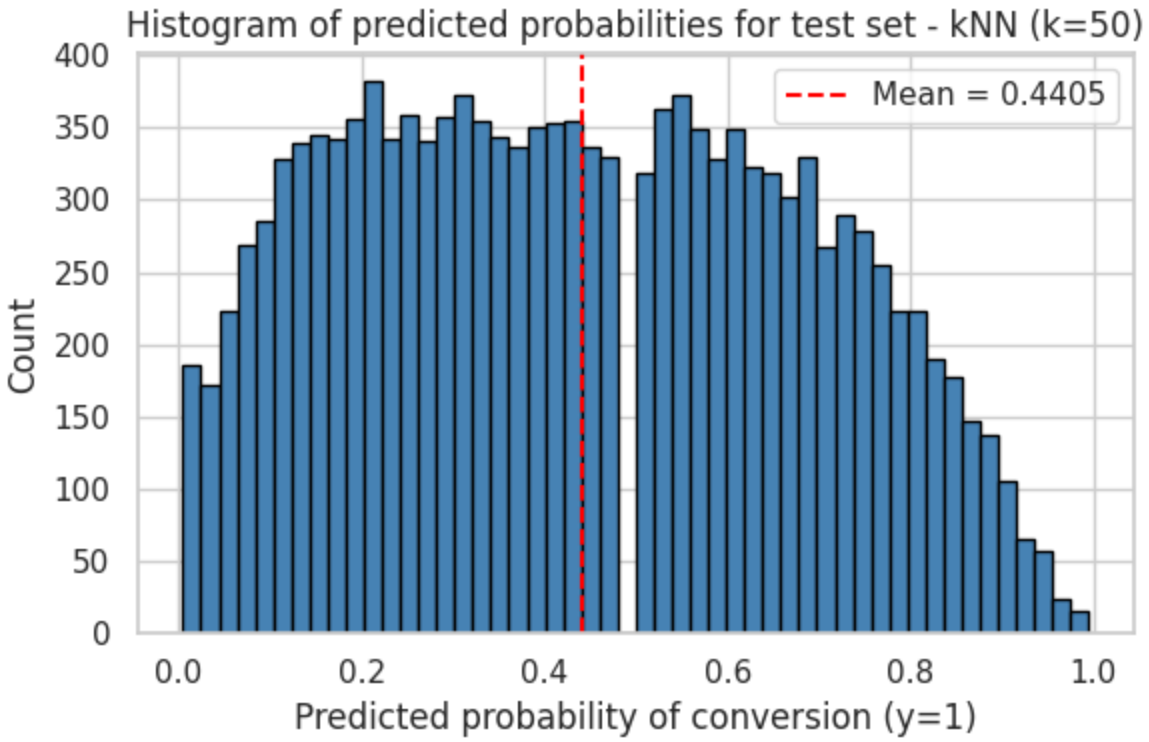
Our confusion matrix here is nearly symettric, kNN doesn't systematically predict one class over another.

Step 4: Generating test predictions

In [116…

```
# refit best model on full training data and predict on test set
print(f"Refitting best model {best_model} on full training data and predi
best_pipe.fit(x_train, y_train)

# Predict probabilities for test set
p_hat = best_pipe.predict_proba(x_test)[:, 1]

# Clip predictions to avoid extreme probabilities that can cause issues i
p_hat = np.clip(p_hat, 0.005, 0.995)

# Sanity checks

print(f"\nNumber of predictions: {len(p_hat)}")
print(f"Min probability: {p_hat.min():.5f}")
print(f"Max probability: {p_hat.max():.5f}")
print(f"Mean predicted probability: {p_hat.mean():.5f}")

fig, ax = plt.subplots(figsize=(6, 4))
ax.hist(p_hat, bins=50, color='steelblue', edgecolor='black')
ax.axvline(p_hat.mean(), color='red', linestyle='--', label=f'Mean = {p_h
ax.set_xlabel('Predicted probability of conversion (y=1)')
ax.set_title(f'Histogram of predicted probabilities for test set - {best_
ax.set_ylabel('Count')
ax.legend()
plt.tight_layout()
plt.show()
```

```
Refitting best model kNN (k=50) on full training data and predicting on te
st set...

Number of predictions: 13564
Min probability: 0.00500
Max probability: 0.99500
Mean predicted probability: 0.44054
```



In [119…

```python
# Create and save submission file
submission = pd.DataFrame({
    'ID': test_ids,
    'p_hat': p_hat
})
submission.to_csv('predictions.csv', index=False)
print("Submission file 'predictions.csv' created with ID and p_hat column
print(f"Submission file shape: {submission.shape}")
print(f"\nfirst 10 rows of submission:\n{submission.head(10)}")
```

```
Submission file 'predictions.csv' created with ID and p_hat columns.
Submission file shape: (13564, 2)

first 10 rows of submission:
   ID  p_hat
0   1   0.66
1   2   0.80
2   3   0.84
3   4   0.50
4   5   0.14
5   6   0.86
6   7   0.36
7   8   0.68
8   9   0.16
9  10   0.30
```